Honors Thesis:

Designing a Universal Information Language

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March 26, 2026

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Computer Science
Outside Reader

Agenda

Introduction

Language Overview

Syntax

Metatheory

Conclusion

Introduction

[1] Ackoff, R. L. (1989). *From Data to Wisdom*. J. Appl. Syst. Anal., 16, 3–9.
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    - Can be a *process*, or *knowledge*
    - Most useful as *thing*
  - **Floridi:** Information is **truthful**[3]

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- **Depth:** domain specific data
  - Terminology, kinds of data, etc.
- **Breadth:** wide variety of domains
  - Different principles, methods
  - Example: Physics $\neq$ Biology!
- **Standardization:** ambiguous interpretations
  - Related issue: being **interoperable**
    - **Interoperable:** a portable data format
    - Example: U.S. Health records
    - Transfer is hard, even for the *same* patient

Existing Solutions

[1] Hitzler et al. (2009). *OWL 2 Web Ontology Language Primer*. W3C.
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- **Resource Description Language (RDF):**[1]
  - ▸ Used in Web Ontology Language **(OWL)**
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  - ▸ Stores **properties** (node/edge metadata)
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  - — No logic engine. Hacky tooling/scripts

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- **Cyc:**[3]
  - ▸ Largest **knowledge base** to date
  - \+ Sophisticated logic engine
  - — Hard-coded, proprietary rules

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  - Old German word, meaning “cloud”, “sky”[1]

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  - Pronounced: “VEL-KIN”
  - Old German word, meaning “cloud”, “sky”[1]
  - Represents *expansive* nature of language

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  - Have a formal specification

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   - Symbol Grounding Problem[2]
   - Treat as free parameter in theory

- Achieve both via **representations**

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- Track *faithfulness* of representations via **truth**

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  - Representations are *also* units

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  - More details later! (Metatheory)

Rules

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Syntax

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  - Encoding: characters
    - Building blocks of strings
  - Grammar: strings allowed by language
    - Punctuation
    - How strings are combined

Encoding

- Characters written in `type-writer font`[1]
- Define US-ASCII: Printable (codes: 32-126) + `EOF` (code: 127)
- Character classes: refer to thesis

[1] Font used: Intel One Mono (https://github.com/intel/intel-one-mono), licensed under OFL 1.1.

US-ASCII

Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph
32	20	Space	46	2E	.	60	3C	<	74	4A	J	88	58	X	102	66	f	116	74	t
33	21	!	47	2F	/	61	3D	=	75	4B	K	89	59	Y	103	67	g	117	75	u
34	22	"	48	30	0	62	3E	>	76	4C	L	90	5A	Z	104	68	h	118	76	v
35	23	#	49	31	1	63	3F	?	77	4D	M	91	5B	[	105	69	i	119	77	w
36	24	$	50	32	2	64	40	@	78	4E	N	92	5C	\	106	6A	j	120	78	x
37	25	%	51	33	3	65	41	A	79	4F	O	93	5D	]	107	6B	k	121	79	y
38	26	&	52	34	4	66	42	B	80	50	P	94	5E	^	108	6C	l	122	7A	z
39	27	'	53	35	5	67	43	C	81	51	Q	95	5F	_	109	6D	m	123	7B	{
40	28	(	54	36	6	68	44	D	82	52	R	96	60	`	110	6E	n	124	7C	|
41	29	)	55	37	7	69	45	E	83	53	S	97	61	a	111	6F	o	125	7D	}
42	2A	*	56	38	8	70	46	F	84	54	T	98	62	b	112	70	p	126	7E	~
43	2B	+	57	39	9	71	47	G	85	55	U	99	63	c	113	71	q			
44	2C	,	58	3A	:	72	48	H	86	56	V	100	64	d	114	72	r			
45	2D	-	59	3B	;	73	49	I	87	57	W	101	65	e	115	73	s			

Table 1: Printable US-ASCII codes and glyphs.

Grammar

- Build on two papers (details in thesis)

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    - Current proofs are done
    - Grammar is **not** finalized

```
start := units - lexeme -> EOF,
units := "" | {unit - lexeme -> *{"",
"," - lexeme -> units}},
unit := node - lexeme -> *{
  ":=" - lexeme -> choices,
  *{"", arc - lexeme -> chain}
},
choices := *{"", "|"} - lexeme ->
choices_list,
choices_list := *{"",
  chain - lexeme -> *{"", "|" - lexeme
-> choice_list}
},
chain := *{"", "*"} - seq -> path -
lexeme -> *{"", arc - lexeme ->
chain},
arc := right_arc | other_arc,
right_arc := "-" - seq -> node - seq -
> "->",
other_arc := "<-" - seq -> node - seq
-> *{"-", "->"},
node := *{"", "*"} - seq -> path,
path := *{"", modifiers} - seq ->
trailer - seq -> *{"", graph},
modifiers := *{
  "@",
  "~" - seq -> *{"", "@"}
},
trailer := segment - seq -> *{"", "."
- seq -> trailer},
segment := HANDLE - lexeme -> *{"",
graph},
graph :=  "{" - lexeme -> units -
lexeme -> "}",
HANDLE := ID | STRING,
STRING := SQ_STRING | DQ_STRING,
SQ_STRING := "'" - seq -> *{SQ_CHAR,
"\'"} - seq_many_till -> "'",
DQ_STRING := '"' - seq -> *{DQ_CHAR,
'\"'} - seq_many_till -> '"',
ID := ID_CHAR - seq_many_till ->
*{RESERVED, WHITESPACE}
```

Listing 7: Full Welkin grammar.

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- Many theorems on limit:s

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Artëmov's Selector Proofs

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Agenda

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Metatheory

Conclusion

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